Deployment environments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask where the run points and what its credential can do.

Teach data, namespace and authority separation with harmless local fixtures.

Keep one habit: Model wrong store → contrast wrong credential → inspect lookup trace → assess development case → record real-isolation limits.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of deployment **environment**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Queues and asynchronous work. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of queues and asynchronous work

Reference example, not the only acceptable wording: Teacher explanation: Queues separate acceptance from completion. Work may be redelivered, so consumers must preserve operation IDs and deduplicate committed effects.

A queued job is pending, running, completed or failed. Acknowledge only after the declared durable completion step. A repeated completed ID returns its prior result; unknown status needs reconciliation.

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of data retention and deletion

Reference example, not the only acceptable wording: Teacher explanation: Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Without opening the old lesson, explain one key idea from External side effects and recovery. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of external side effects and recovery

Reference example, not the only acceptable wording: Teacher explanation: External effects can succeed while their acknowledgement is lost. Recovery must inspect committed state before repeating the operation.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve compensation as a separate authorised action.

Pictured end goal

Explain deployment **environment**s using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Queues and asynchronous work

Data retention and deletion

External side effects and recovery

Use the three-session teaching cycle above, with extra time as needed. Development, staging and production are distinct **environment**s with explicit data and credential boundaries. Test configuration must not silently target production. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of deployment **environment**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach target, namespace and authority as explicit boundaries

> Inspect actual configuration rather than treating a badge as enforcement.

Definitions and complete fictional manifest

Development, staging and production are distinct **environment**s with explicit data and credential boundaries. Test configuration must not silently target production.

For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed **environment** names. A staging run cannot use production write credentials.

Symbolic environment	Store / secret namespace / credential ID	Fictional room
development	dev-fixtures / dev-mock / dev-reader	D
staging	stage-fixtures / stage-mock / stage-reader	Z
production	approved-school / prod-mock / prod-reader	A
Allowed operation	Read only, in the environment and store declared by trusted credential metadata	No writes
Configuration validation	Exact field names; known environment; matching store, namespace and credential authority	Block before fixture lookup on mismatch
Credential IDs	Harmless symbolic labels, not usable keys or secret values	No real secret access

Symbolic environment	Store / secret namespace / credential ID	Fictional room
Isolation limit	Dictionary checks model policy decisions, not actual IAM or network enforcement	No live systems used

Main decision ledger and checked reasoning

Staging request variant	Expected decision	Local read trace
Label staging, store approved-school	Blocked: store crosses environment	Empty
Correct stage-fixtures / stage-mock / stage-reader, read	Allowed fictional rehearsal; returns room Z	stage-fixtures once
Correct staging store but prod-reader	Blocked: wrong credential authority	No added read
Correct staging store but prod-mock namespace	Blocked: wrong secret namespace	No added read
Correct staging fields but write operation	Blocked: read-only authority	No added read
Unknown environment name stage	Blocked; no implicit fallback	No added read

Require separate checks of environment name, target store, secret namespace and trusted credential operation/scope. Main staging-to-production-store configuration blocks before lookup. Correct staging reads fictional Z once; substituting prod-reader blocks even with the correct staging store. Fake data cannot reduce a credential's authority. No configuration silently defaults to production.

The symbolic production dictionary is not live school data. A passing rehearsal supports review rather than automatic promotion, and the code cannot demonstrate real network/IAM separation. The selected development/production-token case transfers the authority check to a different environment but retains the same principle; assess the learner's explanation, not a claim of wholly unfamiliar content.

Deployment environments — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Staging configuration points to production storage while the interface displays staging.	For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed environment names. A staging run cannot use production write credentials.	The label is misleading; block the run through configuration validation. A display badge is not an isolation boundary.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Qualification: Passing staging tests does not prove production limits or operational conditions are identical.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Staging configuration points to production storage while the interface displays staging.

The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Mismatch and representativeness probes

Probe	Expected interpretation
Staging badge but production target	Block; label is not an isolation boundary
Fake records but production-capable credential	Authority crosses environments; fake data does not repair it
Staging store and correct credential, wrong namespace	Block; do not silently borrow production secrets
Unknown environment or missing field	Reject rather than default to production
Read-only credential asked to write	Block operation mismatch
Successful fictional rehearsal	Evidence for supplied fixtures; no automatic promotion
Actual network route/IAM privileges	Not exercised by this symbolic model
Real representative test conditions	Require approved privacy-safe cases, not copied private production data

Separate room-Z rehearsal and changed-credential scope

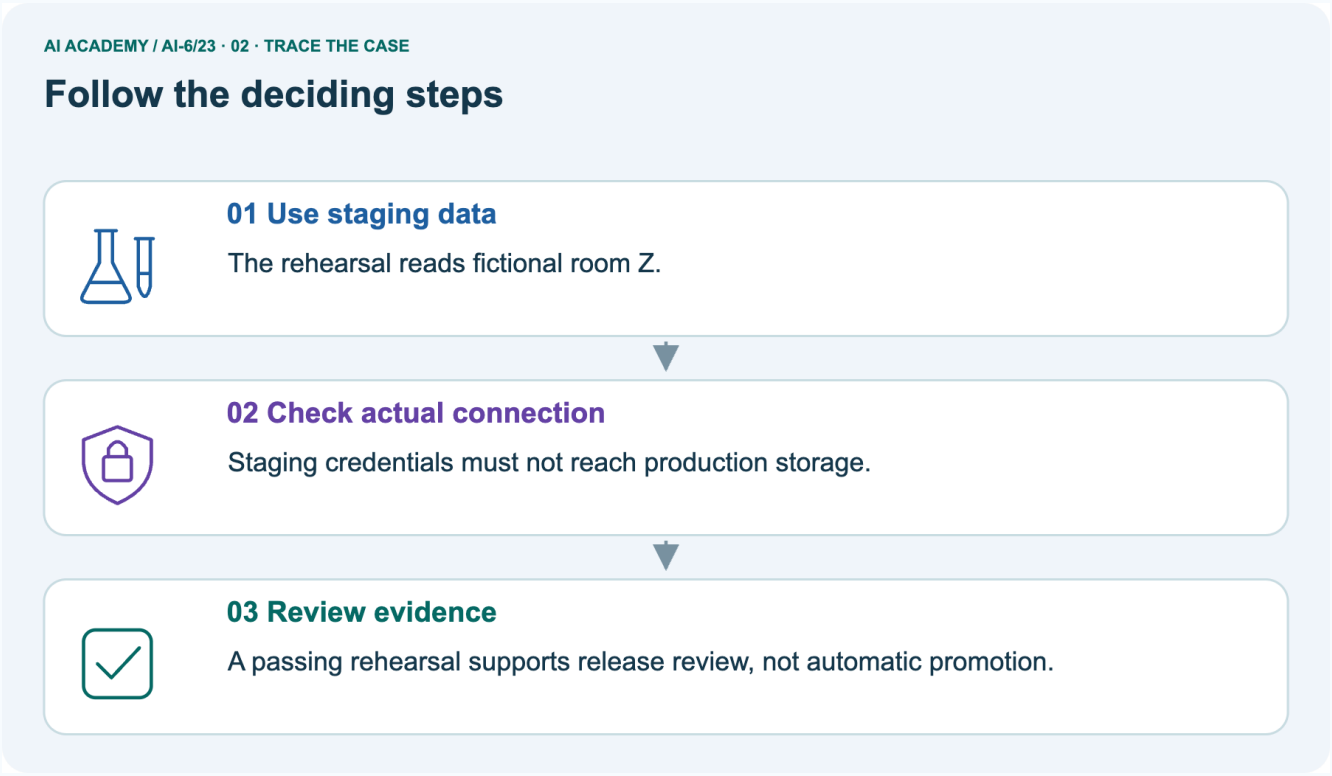
Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records.

Staging card names trial room Z; production card names room A. Credentials and stores are separate.

The picture’s approved rehearsal reads fictional room Z with staging-only authority. Its contrasting production card names room A; in our program both are harmless dictionaries. The main bad configuration points staging at the production-labelled store and must stop before any lookup.

The pictured change instead keeps the staging store but substitutes a production credential; this also blocks, for an independent authority mismatch. A successful room-Z rehearsal supports review of that fixture only. It does not promote code, verify every live edge case or prove actual storage/credential isolation.

Picture trace and boundary key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Use staging data	The rehearsal reads fictional room Z.
2. Check actual connection	Staging credentials must not reach production storage.
3. Review evidence	A passing rehearsal supports release review, not automatic promotion.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An **environment** includes code, configuration, permissions and data. A staging label alone cannot create separation.

Fictional fixtures can miss live edge cases; use approved privacy-safe representative test conditions too.

Change: Point staging at the production credential.

Prediction: Stop the rehearsal because its separation boundary is breached.

Reason: Actual access matters more than the **environment** label.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Environment	A declared deployment/configuration context
Isolation	Preventing unintended cross-context access
Configuration mapping	The relationship between environment and allowed resources

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

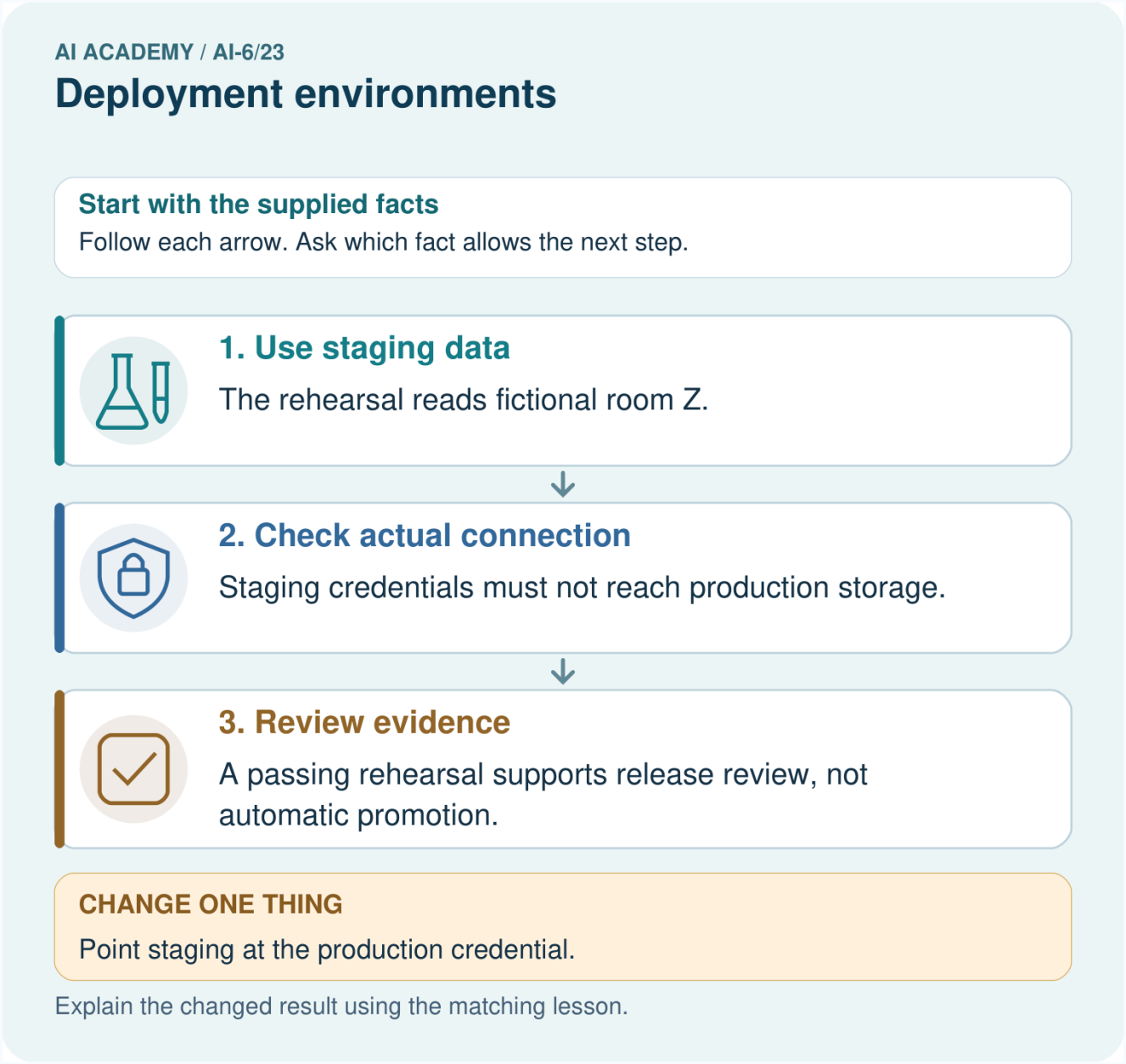
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Deployment environments

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Staging configuration points to production storage while the interface displays staging. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Deployment **environment**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Passing staging tests does not prove production limits or operational conditions are identical.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Identify the staging data and credential boundary. Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

The rehearsal reads fictional room Z. Staging credentials must not reach production storage. A passing rehearsal supports release review, not automatic promotion.

Changed case: Stop the rehearsal because its separation boundary is breached. Actual access matters more than the **environment** label.

Changed-case reasoning: Stop the rehearsal because its separation boundary is breached. Actual access matters more than the **environment** label.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main wrong production store blocks before any lookup. Correct staging reads Z once; wrong production credential or namespace and a write request block without an added read. Selected development fake-record case violates the credential authority boundary: use development-only mock authority and inspect the actual endpoint/store as well as the label.

Passing fixture checks is scoped evidence, not production readiness or real isolation proof.

Diagnose the current misconceptions

The worked result establishes every future case: Passing staging tests does not prove production limits or operational conditions are identical.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Development uses fake records but a production token. What boundary is violated?

Teacher explanation: Credential scope crosses **environment**s. Use development-only mock authority and inspect endpoints as well as labels.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Development, staging and production are distinct **environment**s with explicit data and credential boundaries. Test configuration must not silently target production.

For the classroom mock ENV selects a separate fixture store and secret namespace. Validate allowed **environment** names. A staging run cannot use production write credentials.

W2 Trace the worked case

Staging configuration points to production storage while the interface displays staging. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

W3 Audit the limit

A report omits this qualification: Passing staging tests does not prove production limits or operational conditions are identical.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Passing staging tests does not prove production limits or operational conditions are identical.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Passing staging tests does not prove production limits or operational conditions are identical.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Qualification: Passing staging tests does not prove production limits or operational conditions are identical.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Development uses fake records but a production token. What boundary is violated?

Teacher explanation: Credential scope crosses **environment**s. Use development-only mock authority and inspect endpoints as well as labels.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Credential scope crosses **environment**s. Use development-only mock authority and inspect endpoints as well as labels. Require intermediate reasoning and the stated scope.

Fresh retry: Test **environment** correctly uses fixtures but exports production secrets to logs. Expected reasoning: Secret boundary violated despite test data. Validate **environment**/credential mapping and redact logs.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled deployment **environment**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of deployment **environment**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Release and rollback. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/24, ai-6/26, ai-6/29. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of deployment **environment**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 **environment** before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Staging configuration points to production storage while the interface displays staging. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Check every learner before discussion: Use the worked case for Deployment **environment**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Passing staging tests does not prove production limits or operational conditions are identical.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Point staging at the production credential. Require a **prediction** and a reason before comparing. Expected result: Stop the rehearsal because its separation boundary is breached.

Reason: Actual access matters more than the **environment** label.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

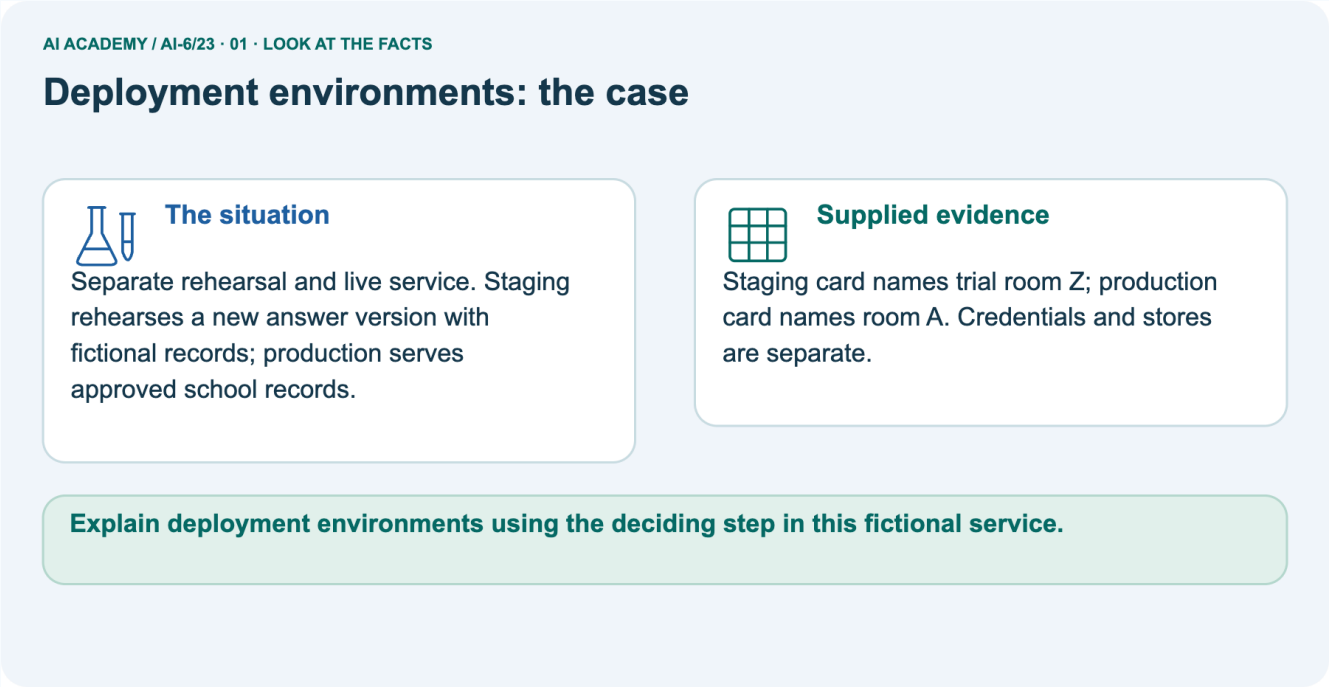
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Deployment environments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

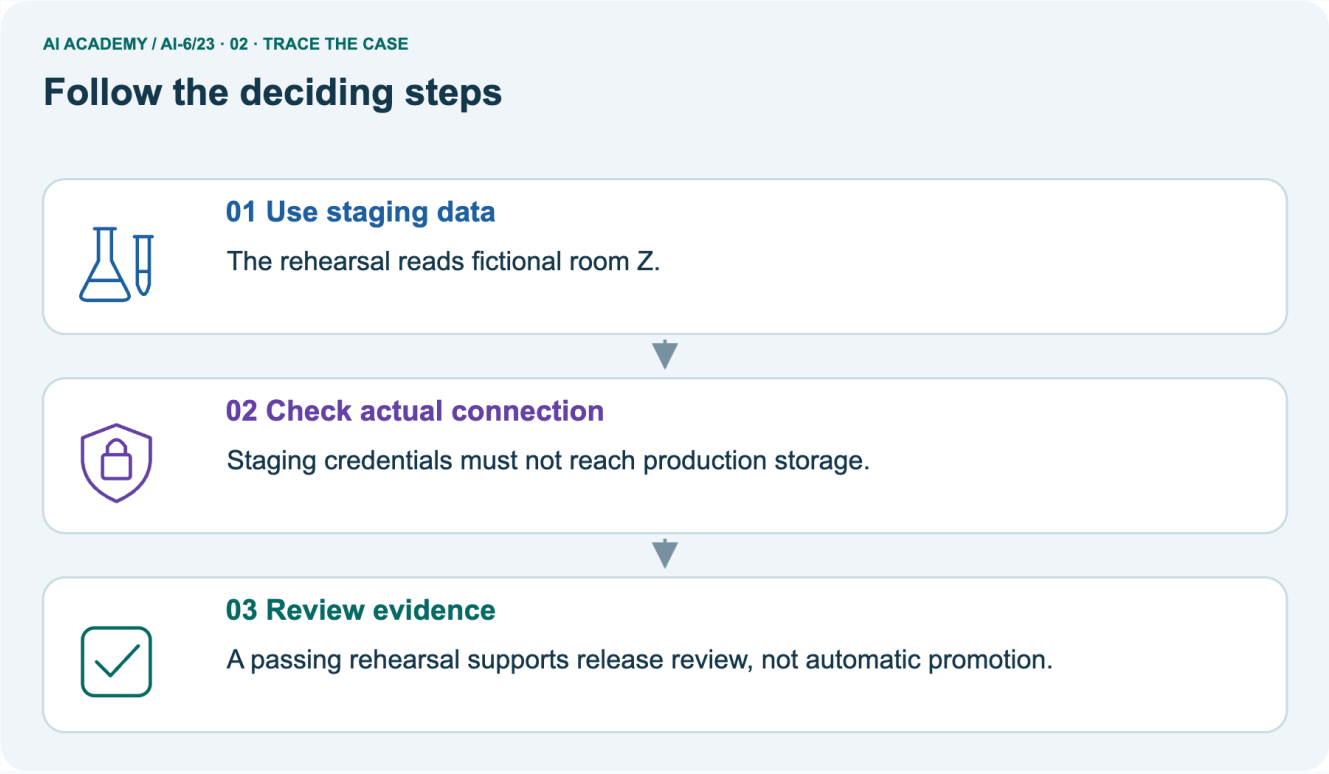
Separate rehearsal and live service. Staging rehearses a new answer version with fictional records; production serves approved school records. Supplied facts: Staging card names trial room Z; production card names room A.

Credentials and stores are separate.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



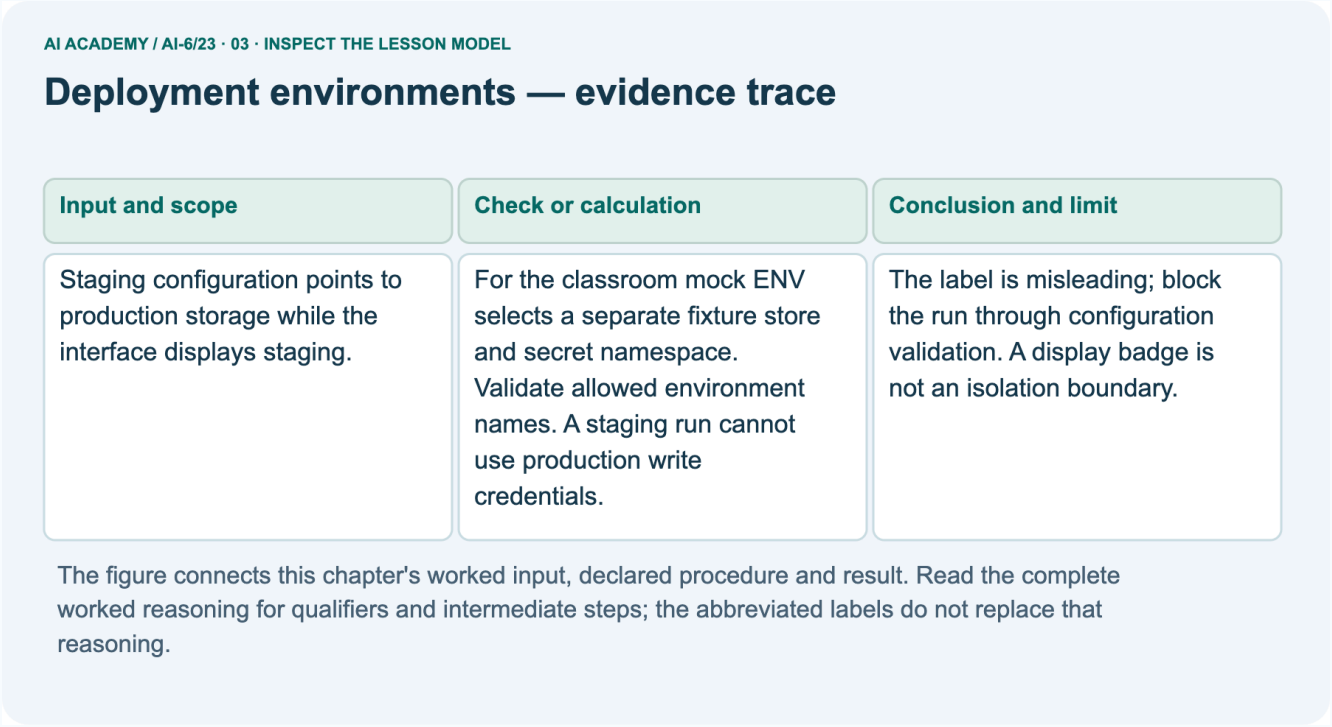
Follow the labelled picture with the matching explanation. Open the vector diagram

Use staging data: The rehearsal reads fictional room Z. → Check actual connection: Staging credentials must not reach production storage. → Review **evidence**: A passing rehearsal supports release review, not automatic promotion.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Deployment environments — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

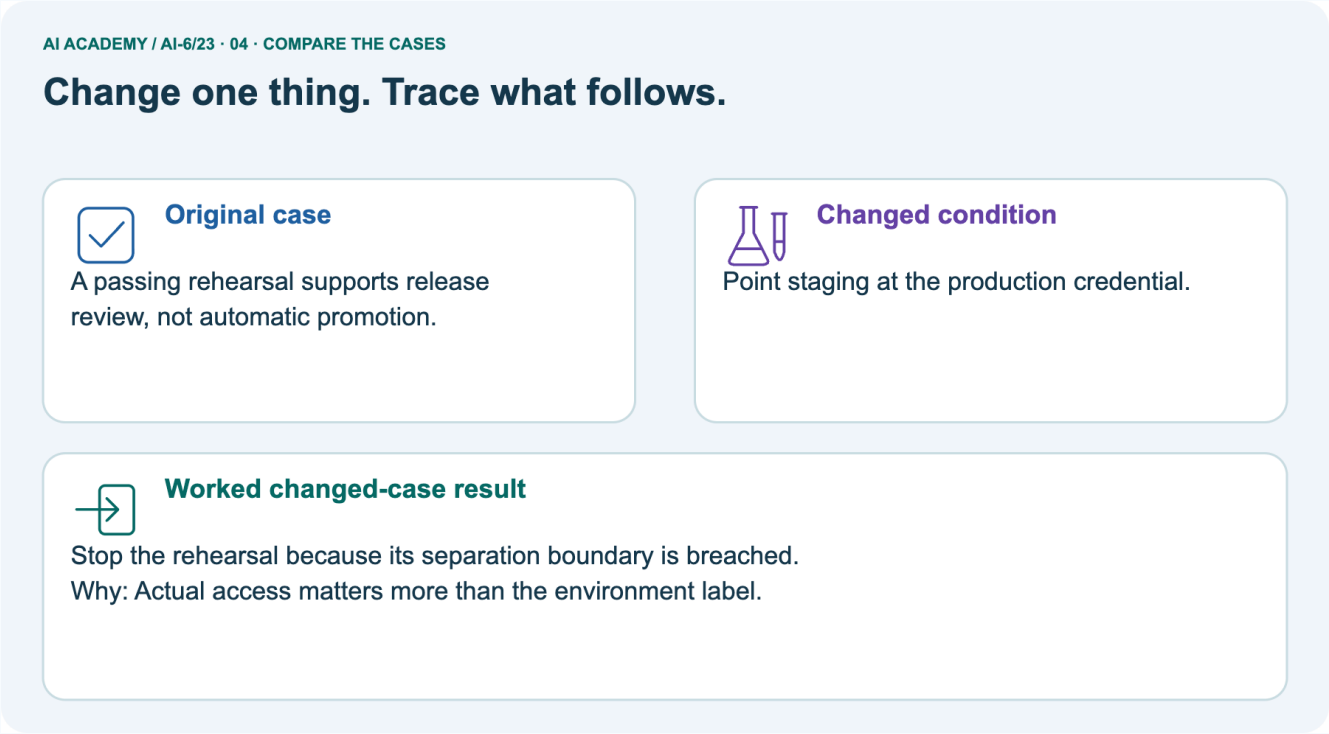
Deployment **environment**s — **evidence** trace. Staging configuration points to production storage while the interface displays staging. | For the classroom mock ENV selects a separate fixture store and secret namespace.

Validate allowed **environment** names. A staging run cannot use production write credentials. | The label is misleading; block the run through configuration validation. A display badge is not an **isolation** boundary.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A passing rehearsal supports release review, not automatic promotion.
Changed condition: Point staging at the production credential.

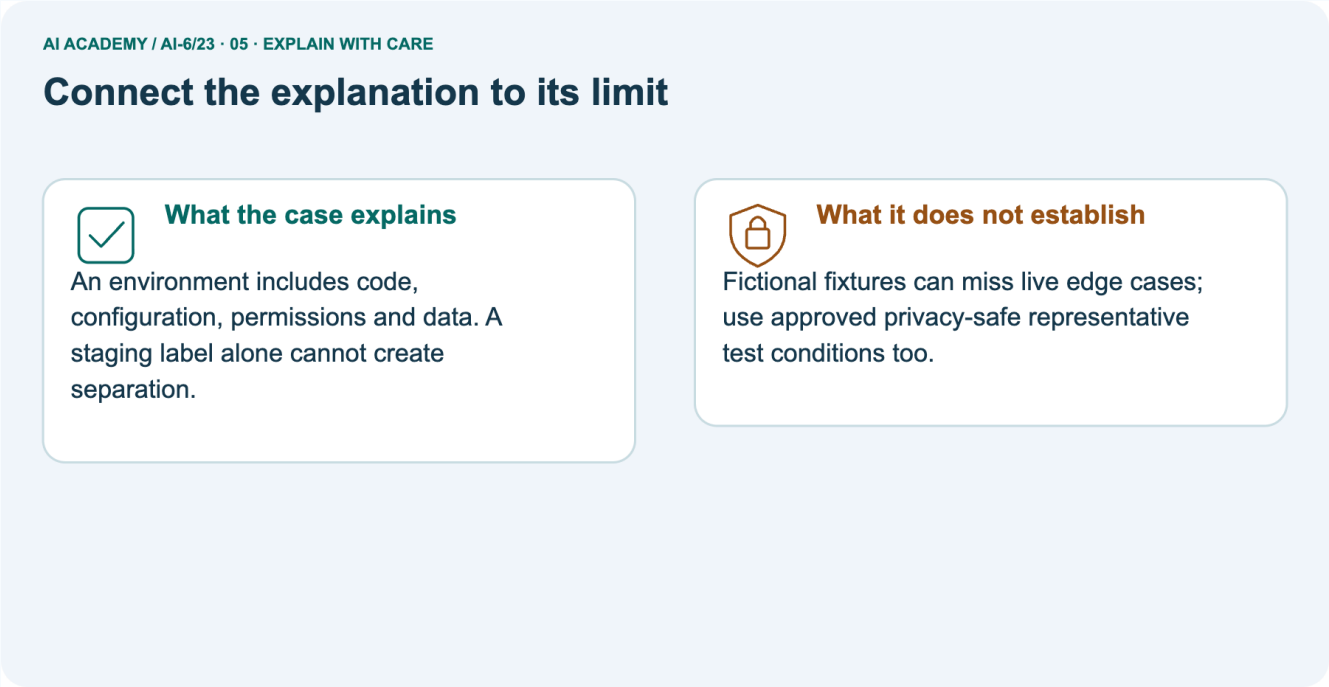
Expected result: Stop the rehearsal because its separation boundary is breached. Reason: Actual access matters more than the **environment** label.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Actual access matters more than the **environment** label.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An **environment** includes code, configuration, permissions and data. A staging label alone cannot create separation.

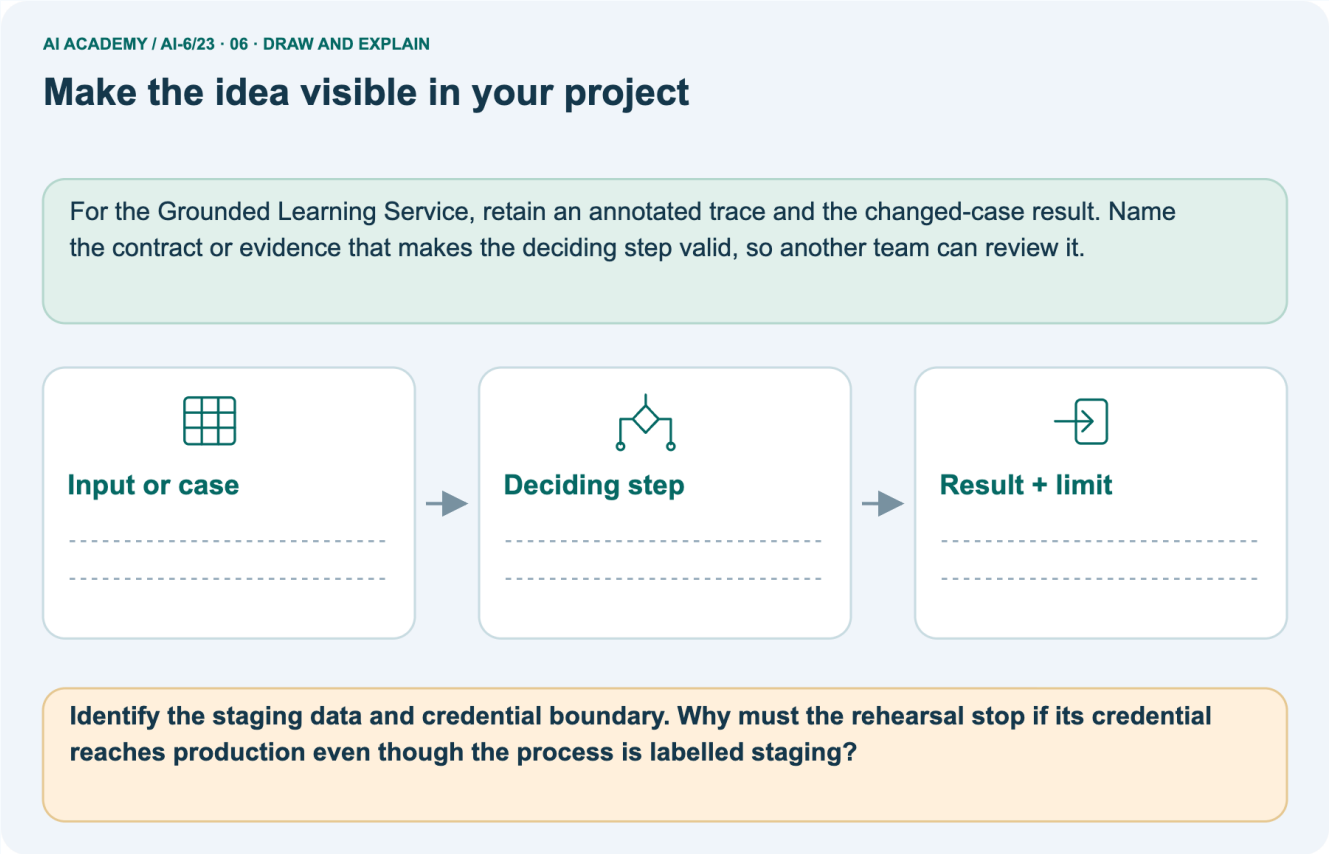
Limit: Fictional fixtures can miss live edge cases; use approved privacy-safe representative test conditions too.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Fictional fixtures can miss live edge cases; use approved privacy-safe representative test conditions too.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The rehearsal reads fictional room Z. Staging credentials must not reach production storage. A passing rehearsal supports release review, not automatic promotion.

Changed case: Stop the rehearsal because its separation boundary is breached. Actual access matters more than the **environment** label.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Deployment environments: the case

02. Follow the deciding steps

03. Deployment environments — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.